

Nicholas D. Gray, Ph.D.

DATA SCIENTIST

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Data scientist with nearly four years of industry experience in insurance analytics, built on a decade of quantitative research training. Combines a Ph.D. in Cognitive Psychology with day-to-day SQL, Python, and applied statistics to turn large, messy data sources into analyses, reports, and recommendations that stakeholders act on. Integrates AI into analysis and reporting workflows where it earns its place, connecting LLMs to existing SQL and Python tooling.

TECHNICAL SKILLS

Languages & Tools	Python (proficient), SQL / Microsoft SQL Server, R, MATLAB, Microsoft Excel (proficient)
Modeling & Statistics	Regression (linear & logistic), XGBoost & gradient boosting (applied), Statistical inference, Experimental design, Model evaluation
Data & Analysis	ETL & data cleaning, Exploratory data analysis, Survey & qualitative analysis
AI & Automation	LLM-assisted analysis & reporting, Automated reporting pipelines, Prompt design, API integration
Communication	Stakeholder reporting & dashboards, Data visualization, Technical writing, Teaching & presentation

PROFESSIONAL EXPERIENCE

Data Scientist

Oct 2022 – Present

Safety National Casualty Corporation · St. Louis, MO

- Build analytics reports and dashboards used by internal teams and external stakeholders across the business.
- Built an automated benchmarking report that calls an LLM to generate written insights from the underlying data, integrating SQL extraction, Python processing, and generated narrative into a single pipeline.
- Extract and join data from large, complex sources in Microsoft SQL Server, then transform and clean it in Python for analysis.
- Run and adjust established machine-learning models (XGBoost), validating output and interpreting predictions for underwriting and claims questions.
- Design custom analyses out of exploratory investigation, tailoring method and reporting to the decision each stakeholder actually faces.

Postdoctoral Researcher — Quantitative Analysis

Aug 2019 – Oct 2022

Institute for Successful Longevity, Florida State University · Tallahassee, FL

- Led statistical analysis for large-scale behavioral clinical trials on healthy cognitive aging, working across longitudinal datasets and multiple analysis platforms.
- Established statistical performance norms and built regression models identifying the variables that drive cognitive outcomes.
- Integrated qualitative streams — surveys and interviews — with quantitative measures to produce findings neither would support alone.
- Designed and programmed novel studies end to end and published the results in peer-reviewed journals.

Graduate Researcher

Aug 2013 – May 2019

Department of Psychology, Florida State University · Tallahassee, FL

- Modeled experimental data with logistic regression and related statistical methods across several software platforms.
- Designed, programmed, and ran grant-funded experimental research; managed the lab, recruiting and supervising undergraduate researchers.
- Taught laboratory sections of Cognitive Psychology for three years and co-instructed an online Careers in Psychology course, translating research concepts and methods for non-expert audiences.

SELECTED PROJECTS

adherence – Python · github.com/ndg13b/adherence

Scores how regular a behavior is — not just how often it happens — from nothing but a list of timestamps, then forecasts when the next session will land. The same model applies to medication timing, exercise, physiotherapy, or a training program.

CivicGateway – JavaScript · civicgateway.org

A nonpartisan site showing upcoming elections and elected officials by city; static front end over a Supabase/PostgreSQL back end with a schema designed from scratch, row-level security, and an offline fallback.

tuned-in – JavaScript · ndg13b.github.io/tuned-in

A teaching collection pairing pop-culture clips with the cognitive psychology they demonstrate — and, just as often, get wrong. Each concept is labeled accurate, overstated, or myth.

EDUCATION

Ph.D., Cognitive Psychology

2019

Florida State University

Thesis: The causes of and influences on being reminded

M.S., Cognitive Psychology

2016

Florida State University

Thesis: Avoiding interference: Differentiation and reminding

B.S., Psychology, cum laude

2012

Ohio State University

Certification: Machine Learning Specialization — DeepLearning.AI & Stanford Online (Coursera), 2026. Supervised learning (regression, classification, neural networks, decision trees), unsupervised learning, recommender systems, reinforcement learning

Graduate methods training: ANOVA, regression, meta-analysis, structural equation modeling

Additional coursework: Machine Learning with Python; Microsoft SQL Server (independently completed)

SELECTED PUBLICATIONS

Gray, N., & Charness, N. (2022). Technology obsolescence across the adult lifespan in a USA internet sample. *Frontiers in Public Health*, 10, 1005822.

Gray, N., Yoon, J.-S., Charness, N., Boot, W. R., et al. (2021). Relative effectiveness of general versus specific cognitive training for aging adults. *Psychology and Aging*, 37(2), 210–221.

Carr, D. C., Tian, S., He, Z., et al., incl. Gray, N. (2022). Motivation to engage in aging research: Are there typologies and predictors? *The Gerontologist*, gnac035.

Charness, N., Boot, W. R., & Gray, N. (2020). Mobile monitoring and intervention technology (MMI) for adaptive aging. In *Mobile Technology for Adaptive Aging: Proceedings of a Workshop* (pp. 21–40). National Academies Press.

Conference presentations at the Psychonomic Society (2018) and the Cognitive Aging Conference (2022).